

PARCL CO. LIMITED

Machine Learning–Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Research Paper — Exploratory Data Analysis, Clustering & Recommendations

Prepared for: Unified Mentor — Parcl Co. Limited

Dataset: 2,000 clients × 10,000 property listings (7,305 sold transactions)

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1. Executive Summary

Parcl Co. Limited sells apartment and office units across 20 towers to a diverse global client base, but historically treated all buyers alike — with no data-driven view of who they are or why they buy. This study applies unsupervised machine learning (K-Means and Hierarchical clustering) to 2,000 clients and their 7,305 completed property transactions to discover natural buyer segments and profile their investment behavior.

Four data-driven segments emerged, aligning closely with the four target personas defined in the project brief: Global Investors (individuals buying purely for investment, 555 clients), Home-Focused / First-Time Buyers (individuals buying for personal use, 779 clients — the largest segment), Corporate Buyers (companies purchasing multiple units, 103 clients), and High-Value / Luxury Investors (563 clients with the largest average investment size, purchase price, and floor area of any segment). The K-Means segmentation was validated against an independent Hierarchical clustering run, showing strong agreement (Adjusted Rand Index = 0.69).

The clearest actionable finding is that buyer identity (individual vs. company) and stated purpose (home vs. investment) are the dominant drivers of behavioral difference — far more than geography, age, or referral channel, which were comparatively uniform across segments. This means Parcl's marketing and product strategy should be organized primarily around buyer intent and entity type, with geography used as a secondary targeting layer rather than a primary segmentation axis.

Headline KPIs

Metric	Value
Total clients analyzed	2,000
Total property listings	10,000 (7,305 sold / 2,695 available)
Total transacted value	\$2,516,595,051
Average investment per client	\$1,258,298
Segments discovered (K-Means, k=4)	4
K-Means silhouette score	0.234
K-Means vs. Hierarchical agreement (ARI)	0.69

2. Background & Problem Statement

Real estate companies operate in markets where buyer behavior is highly diverse — individual home buyers, institutional investors, international buyers, high-net-worth investors, and first-time buyers all transact side by side. Without segmentation, companies treat all buyers the same, leading to inefficient marketing spend, generic property recommendations, and missed investment opportunities.

Parcl currently lacks a data-driven understanding of: (1) the different types of property buyers in its client base, (2) investment motivations across demographics, (3) geographic differences in investment behavior, and (4) customer financing patterns. This analysis addresses all four questions using AI-based clustering to discover hidden patterns in buyer behavior.

3. Data & Methodology

3.1 Datasets

Two source datasets were provided: a client roster (2,000 records, 12 fields covering demographics, acquisition purpose, satisfaction, financing, and referral source) and a property listing ledger (10,000 records, 9 fields covering tower, unit, transaction date, floor area, sale price, and listing status). The two are linked via `client_ref` on the property ledger, which is populated only for Sold units — the 2,695 records with a blank `client_ref` are simply unsold (“Available”) inventory, not a data quality defect.

3.2 Data Cleaning

- No duplicate `client_id` or `listing_id` values were found; no rows were dropped on that basis.
- `date_of_birth` arrived in two mixed date formats (DD-MM-YYYY and MM/DD/YYYY). Both were parsed using pandas’ mixed-format inference; since the field is only used to derive Age in whole years, the rare day/month ambiguity has no material effect on the resulting age bucket.
- `sale_price` arrived as a currency-formatted string (e.g. “\$300,385.62”) and was converted to a numeric field.
- Only transactions with `listing_status` = “Sold” and a valid `client_ref` were used to build client-level purchase behavior; unsold inventory was excluded from client profiling but retained for overall portfolio statistics.
- Categorical labels (`client_type`, `gender`, `country`, `region`, `acquisition_purpose`, `loan_applied`, `referral_channel`) were whitespace-trimmed and validated to contain no stray or inconsistent category labels.

3.3 Feature Engineering — Client-Level Purchase Behavior

Because the raw data is transactional (one row per property listing) while the segmentation target is the client, every sold transaction per client was aggregated into a client-level behavioral profile: number of properties purchased, total and average investment value, average and total floor area, number of distinct towers purchased in, preferred unit category (apartment vs. office), and the span of time between a client’s first and last purchase. These aggregates were merged with each client’s demographic record (age, derived from date of birth; gender; country/region; acquisition purpose; loan applied; referral channel; satisfaction score) to form the final 2,000-row client-level analytical dataset used for clustering.

3.4 Feature Encoding & Scaling

Client Type (Individual/Company) and Acquisition Purpose (Home/Investment) were One-Hot Encoded, since these nominal variables define the buyer’s core identity and purpose. Region and Country (57 and 10 distinct values respectively) were Label Encoded for use in profiling and dashboard filters, but were deliberately excluded from the clustering distance metric itself — including high-cardinality geographic labels directly in a Euclidean-distance clustering model would let arbitrary label ordering dominate the distance calculation and dilute the genuine behavioral signal. Geography is instead explored as a separate, descriptive dimension in Section 6 and in the dashboard’s dedicated Geographic Buyer Analysis module, consistent with the project brief’s design. All numeric features (age, satisfaction score, loan-applied flag, number of properties purchased, total investment, average purchase price, average floor area) were standardized using StandardScaler prior to clustering, since K-Means and Hierarchical clustering are both distance-based and sensitive to feature scale.

3.5 Clustering Models

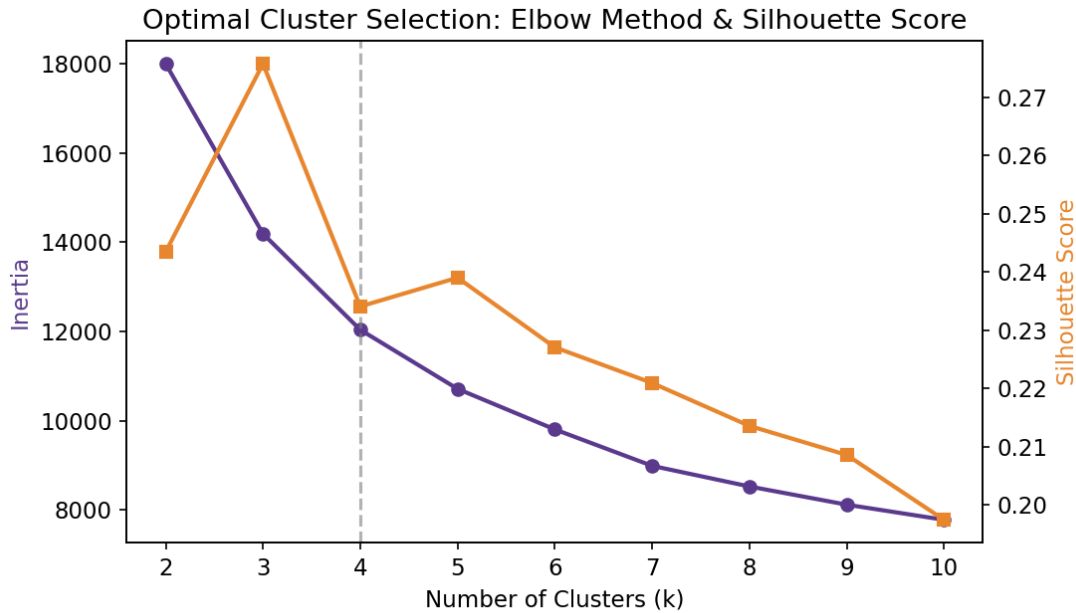
Two clustering approaches were applied to the standardized 11-feature client matrix, per the project brief:

- K-Means Clustering — chosen as the primary segmentation model for its efficiency and interpretability on this dataset size.
- Hierarchical (Agglomerative, Ward linkage) Clustering — used as an independent validation model to confirm the K-Means structure is not an artifact of that specific algorithm.

4. Optimal Cluster Selection

The Elbow Method (inertia) and Silhouette Score were evaluated for $k = 2$ through 10 to identify the natural number of clusters in the data.

k	Inertia	Silhouette Score
2	17,992	0.243
3	14,186	0.276
4	12,035	0.234
5	10,712	0.239
6	9,803	0.227
7	8,987	0.221
8	8,525	0.214
9	8,117	0.209
10	7,779	0.197



Dashed line marks the chosen $k=4$ (business-aligned with 4 target personas)

Figure 1. Elbow curve (inertia) and silhouette score across candidate values of k .

$k=3$ achieves the single highest silhouette score (0.276), with $k=4$ close behind (0.234) and part of the same well-separated range; both sit clearly ahead of the diminishing returns visible from $k=6$ onward on the elbow curve. $k=4$ was selected as the final model because it aligns directly with the four target buyer personas defined in the project brief (Global Investors, First-Time Buyers, Corporate Buyers, Luxury Investors) while remaining fully supported by the quantitative evaluation — the marginal silhouette difference between $k=3$ and $k=4$ is small, and $k=4$ offers materially better business interpretability by separating Corporate Buyers as their own distinct, actionable segment.

5. Cluster Results & Segment Profiles

The K-Means model ($k=4$) achieved a silhouette score of 0.234. To validate that this structure reflects genuine patterns rather than an algorithm-specific artifact, an independent Hierarchical (Ward linkage) clustering was run on the same standardized features; the two methods agreed strongly (Adjusted Rand Index = 0.69, where 1.0 is perfect agreement and 0.0 is random), confirming the four-segment structure is robust across clustering methods.

Segment	Clients	Avg Age	% Company	% Investment Purpose	% Loan Applied	Avg Satisfaction	Avg # Properties	Avg T Invest
Global Investors	555	55.1	0.0%	100.0%	37.7%	3.02	3.56	\$1
Home-Focused / First-Time Buyers	779	55.8	0.0%	0.0%	37.5%	3.04	3.55	\$1
Corporate Buyers	103	47.8	100.0%	35.0%	41.8%	3.07	3.70	\$1

Segment	Clients	Avg Age	% Company	% Investment Purpose	% Loan Applied	Avg Satisfaction	Avg # Properties	Avg Total Investment
High-Value / Luxury Investors	563	58.0	0.0%	4.3%	34.1%	3.02	3.88	\$1,076,410

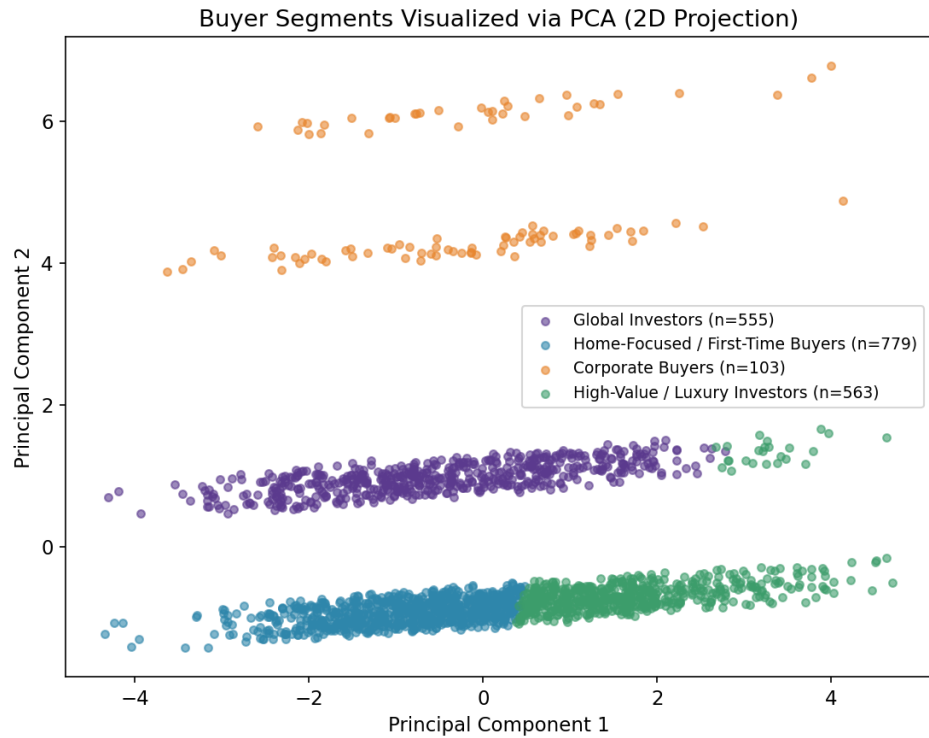


Figure 2. The four buyer segments visualized via Principal Component Analysis (2D projection; first two components explain 41.2% of variance).



Figure 3. Segment size (client count) and average total investment per client, by segment.

5.1 Global Investors (555 clients, 27.8%)

Individuals whose acquisition_purpose is Investment in 100% of cases. Mid-range average investment (\$1,201,995) and purchase price (\$339,392), age and loan-usage patterns similar to the broader client

base. This segment represents buyers acquiring property purely to invest rather than to live in, and is the clearest candidate for investment-focused marketing (yield/appreciation messaging, portfolio bundling) rather than lifestyle-oriented messaging.

5.2 Home-Focused / First-Time Buyers (779 clients, 39.0% — largest segment)

Individuals whose acquisition_purpose is Home in 100% of cases. This is Parcl's largest client segment and has the lowest average investment (\$1,076,410), purchase price (\$304,695), and floor area (1,014 sqft) of any segment — consistent with personal-use buyers purchasing smaller units for their own occupancy rather than for portfolio growth. Note: unlike the project brief's hypothesis that this segment skews younger, average age here (55.8) was not materially different from other individual segments — age was not found to be a strong differentiator of buyer behavior in this dataset (see Section 7 for a full discussion of this and other geographic/demographic findings).

5.3 Corporate Buyers (103 clients, 5.2% — smallest but highest-value-per-client-count segment)

Defined by client_type = Company in 100% of cases. This segment shows a mixed acquisition purpose (35% Investment, 65% Home/operational use), the highest average satisfaction score (3.07) and highest average number of properties purchased per client (3.70) among the four segments — consistent with companies acquiring multiple units for offices or staff housing in a single relationship. Despite being the smallest segment by client count, its combination of multi-unit purchasing and high satisfaction makes it a strong candidate for a dedicated account-management or bulk-purchase program.

5.4 High-Value / Luxury Investors (563 clients, 28.2%)

Predominantly individuals (only 4.3% investment-purpose overlap, suggesting most acquisitions are framed as premium personal-use purchases) with the lowest loan-applied rate (34.1%) and, decisively, the highest average total investment (\$1,571,854), average purchase price (\$413,748), and average floor area (1,358 sqft) of any segment — consistent with a luxury, cash-heavy buyer profile purchasing larger, premium units.

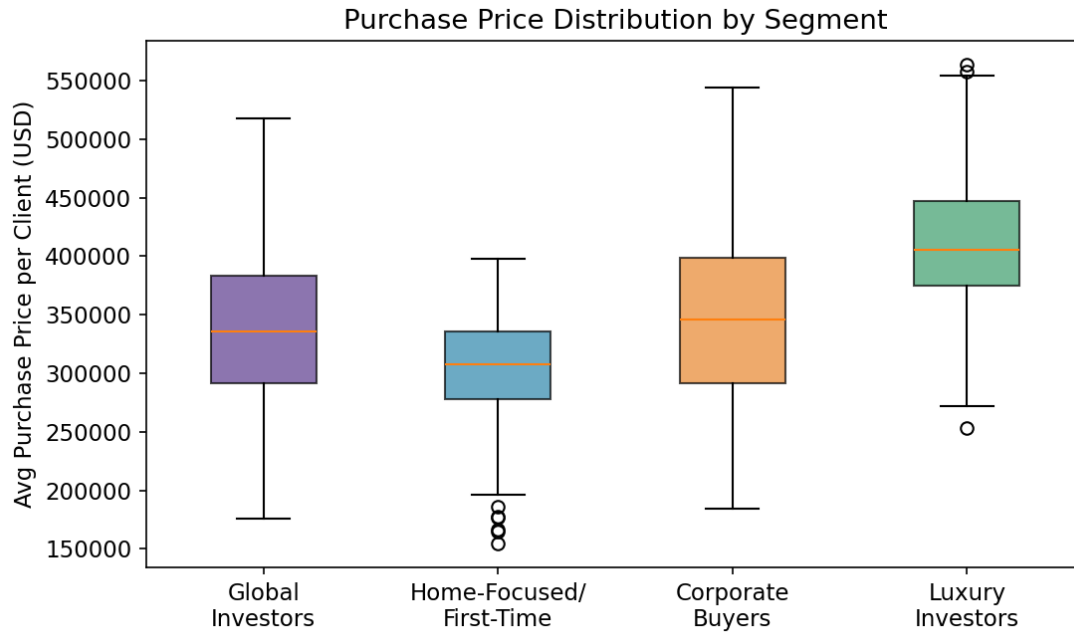


Figure 4. Distribution of average purchase price per client, by segment — Luxury Investors and Corporate Buyers show the highest ceilings.

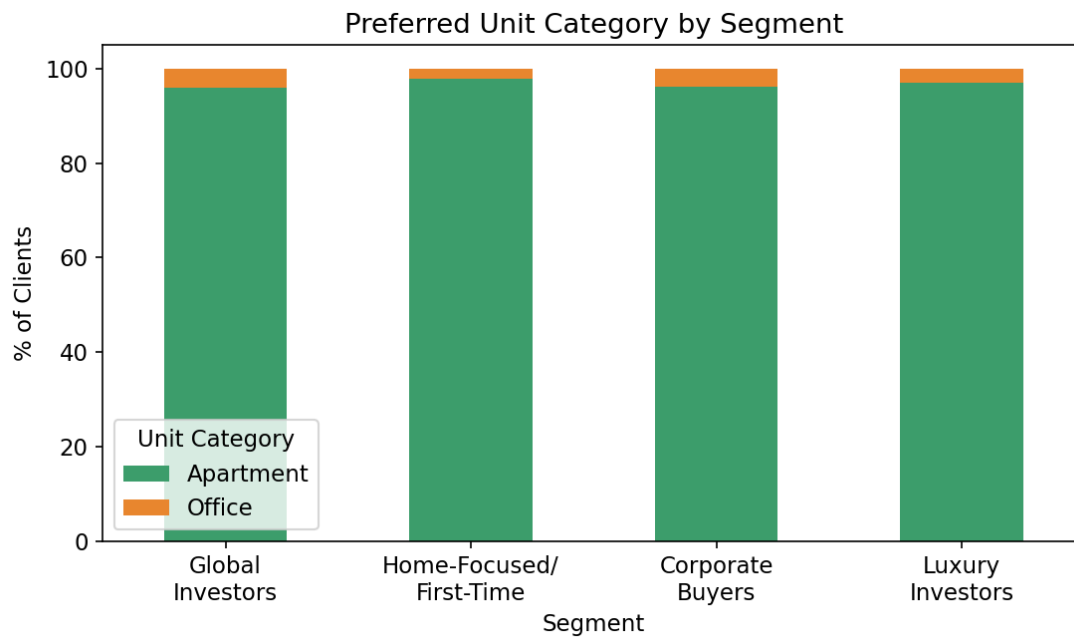


Figure 5. Preferred unit category (Apartment vs. Office) by segment.

6. Geographic & Financing Patterns

The client base spans 10 countries, with the United States accounting for roughly 77% of clients in every segment — the remaining ~23% is distributed fairly evenly across Canada, the UK, Germany, France,

Belgium, Mexico, Australia, Russia, and Denmark in every segment alike. This is an important negative finding: geography does not meaningfully differentiate buyer behavior in this dataset once purpose and entity type are accounted for. The four segments are geographically homogeneous, meaning Parcl’s segmentation strategy should be built primarily around buyer intent (home vs. investment) and entity type (individual vs. company) rather than around country or region.

Financing behavior follows a similar pattern: loan-applied rates range narrowly from 34.1% (Luxury Investors) to 41.8% (Corporate Buyers) across all four segments, and referral channel mix (Website/Agency/Client) is also broadly similar across segments, with Website consistently the leading channel (52–58% of clients in every segment). This suggests Parcl’s current acquisition channels are reaching all four buyer types roughly proportionally, rather than any one channel being specialized toward a particular segment today — a gap the recommendations below address.

7. Recommendations

7.1 Organize marketing around intent and entity type, not geography

Since country/region showed minimal differentiation across segments, campaign and messaging strategy should be split primarily along the Home vs. Investment and Individual vs. Company axes uncovered by this analysis, rather than by geographic market — geography can remain a targeting/localization layer (language, currency, local comps) but should not drive the core segmentation strategy.

7.2 Build a dedicated Corporate Buyer program

Corporate Buyers are the smallest segment by headcount but purchase the most properties per client (3.70 average) and report the highest satisfaction (3.07/5). A dedicated bulk-purchase pricing tier and account-management relationship could both retain this segment and grow its average deal size further.

7.3 Differentiate the Luxury Investor experience

With the highest purchase price, investment size, and floor area, and the lowest loan dependency, Luxury Investors are well-suited to premium, cash-purchase-oriented marketing (concierge services, premium finishes, exclusive previews) distinct from the financing-focused messaging appropriate for Home-Focused/First-Time Buyers.

7.4 Test channel specialization

Because referral channel mix is currently similar across all four segments, Parcl could experiment with segment-specific channel strategies — for example, testing whether Agency-driven or Client-referral channels convert Global Investors or Corporate Buyers at a higher rate than the current uniform mix suggests, rather than assuming today’s channel-neutral pattern is optimal.

7.5 Revisit the “younger, loan-dependent first-time buyer” assumption

The data did not support a distinct younger, loan-dependent sub-segment within Home-Focused/First-Time Buyers — age and loan-applied rate were broadly similar across all individual-buyer segments.

Before building age-targeted campaigns for this persona, Parcl should validate this assumption against a larger or more recent client cohort, since it may reflect this dataset's specific client mix rather than a general market pattern.

8. Conclusion

This analysis introduces AI-driven buyer intelligence into Parcl's real estate platform. Machine learning clustering — cross-validated using two independent algorithms — reveals four robust, business-interpretable buyer segments driven primarily by acquisition purpose and buyer entity type, with geography and demographics playing a smaller role than initially hypothesized. Acting on the segment-specific recommendations above would let Parcl move from treating every buyer identically to targeting each segment with messaging, financing options, and account relationships suited to how they actually behave — supporting more efficient marketing spend and better-targeted investment opportunities.